

SAM ZARRINRAFIE

Electrical Engineering | Robotics & Embedded Systems | AI Integration

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PROFESSIONAL PROFILE

Electrical Engineering student at York University and Robotics and Electronics Lead Engineer at STEM MINDS CORP. Developed Farmtron, an automated farming system, with work spanning PCB design, embedded programming, mechanical prototyping, Linux software, and AI integration. Research interests: radar, wireless systems, robotics, and practical applications of AI.

CORE TECHNICAL SKILLS

Embedded & Programming

Arduino, ESP32, STM32 integration, C/C++, Python, Linux, Tkinter, serial communication, MATLAB, basic FreeRTOS.

Electronics & PCB

Altium Designer, custom PCB design, LTspice, PSpice, sensors, motors, servos, stepper drivers, oscilloscopes, soldering.

AI & Connected Systems

LLM APIs, tool calling, speech recognition, text-to-speech, REST/JSON, Cloudflare, remote control, Git/GitHub.

CAD & Prototyping

Fusion 360, SolidWorks, 3D printing, CNC/CAM, CoreXY kinematics, hardware debugging, technical documentation.

PROFESSIONAL EXPERIENCE

Robotics and Electronics Lead Engineer | STEM MINDS CORP

June 2025 - Present | Aurora, Ontario

- Developed Farmtron from prototype hardware through to working control software, with responsibility for electronics, embedded programming, and system integration.
- Designed and built prototype components using CAD, 3D printing, and PCB design; wired and tested sensors, motors, and control electronics.
- Developed features including AI and voice control, automatic tool docking, a custom Linux interface, and remote operation.
- Troubleshoot mechanical, electrical, and software issues through hands-on testing. Wrote assembly instructions and technical documentation to support product development.

EDUCATION

Bachelor of Engineering, Electrical Engineering | York University

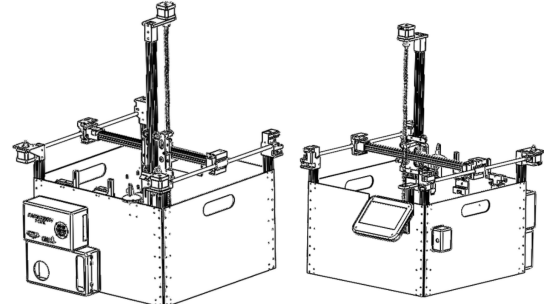
Expected April 2028 | Toronto, Ontario

Academic focus: electrical circuits, programming, engineering design, mathematics, statistics, and systems analysis.

Farmtron | Automated CoreXY Farming System

Product development at STEM MINDS CORP

Python, C/C++, Arduino UNO Q, Linux, custom PCB, sensors, CAD, Cloudflare



- **Motion and tools:** Built a CoreXY gantry with a vertical toolhead for soil probing, watering, and automatic tool changes. Integrated stepper motors, pump control, sensors, homing, and emergency stops.
- **Custom PCB:** Designed a PCB for the system and connected the control electronics, sensors, and actuators. Tested the electronics with the firmware and mechanical assembly.
- **AI and voice control:** Integrated an LLM and speech recognition so users could issue natural-language commands. Connected the assistant to machine-control functions and displayed responses and machine status in the interface.
- **Optical docking:** Developed an automatic tool-alignment system using two light sensors. Compared readings with a tracking LED on and off to isolate its signal, and added alignment scans, live sensor graphs, and simulation tools for testing.
- **Custom Linux environment:** Built a Debian-based system on Arduino UNO Q with a touchscreen interface, networking, and AI services. Kept motor timing, endstop monitoring, and emergency-stop handling in the microcontroller firmware.
- **Remote control:** Developed a Cloudflare-hosted service for device pairing, authenticated commands, and remote status monitoring. Added signed update manifests, file integrity checks, and support for reverting software updates.
- **Testing and refinement:** Developed map-based controls and coordinated motion commands. Used simulation and physical testing to debug homing, tool alignment, motor drivers, and the control interface.

Wireless Robotic Arm | Remote Control & Embedded Prototyping

Arduino, servo control, wireless communication, onboard power, CAD, 3D printing

- Built a multi-axis robotic arm with wireless remote control, servo-driven joints, and onboard power.
- Designed the arm assemblies in CAD and produced 3D-printed parts. Adjusted the layout and fit of components to accommodate the servos, electronics, and wiring.
- Connected wireless commands to Arduino-based servo control and assembled the wiring and power connections needed to operate the arm.
- Tested joint movement, power delivery, and mechanical fit. Troubleshoot wiring and assembly problems and refined the prototype through repeated testing.